THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** push or a pull	Forces are pushes or pulls.
2	**(d)** force acting per unit area	P = F / A.
3	**(a)** tiny particles	Matter consists of very small particles.
4	**(c)** element	That is the definition of an element.
5	**(a)** solution	Uniform composition.
6	**(c)** rough surfaces	Rough surfaces interlock.
7	**(c)** barometer	Barometer.
8	**(a)** melting	Fusion.
9	**(d)** malleability	Malleable.
10	**(b)** brass	Alloy.
11	**(a)** positive and negative	Two kinds of charge.
12	**(a)** very low pressure	Low pressure centre.
13	**(c)** increase in temperature	Warmer liquids evaporate faster.
14	**(a)** a combination of two or more substances not chemically combined	Components keep properties.
15	**(a)** (mass of solute / mass of solution) x 100	Standard definition.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	A push or a pull (1); Newton (N) (1).	1+1
17	Force per unit area; P = F/A (1); SI unit pascal (Pa) (1).	1+1
18	Matter is made of tiny particles (1); Particles move and have spaces between them (1).	1+1
19	Pure substance made of only one kind of atom (1); e.g. iron, oxygen (1).	1+1
20	Solute is dissolved, solvent dissolves it (1); Solution is their homogeneous mixture (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Force can squeeze, stretch or bend (1); Pressing dough, stretching a rubber band (1); Shape change depends on force and material (1).	1+1+1
22	P = F / A (1); = 600 / 0.5 (1); = 1200 Pa (1).	1+1+1
23	Dissolve a spoon of salt/sugar in a beaker of water (1); Level of water hardly rises (1); Solute particles fit in the gaps between water particles (1).	1+1+1
24	Lustre: gold (1); Malleability: aluminium foil (1); Conductivity: copper wire (1).	1+1+1
	OR: Add salt/sugar in portions to 100 mL of cold and of warm water (1); Count spoons dissolved till saturation (1); More dissolves in warm water (1).	1+1+1
25	A metal rod with two thin leaves in a jar (1); Touching a charged body spreads charge to leaves (1); Like charges repel; leaves diverge (1).	1+1+1
	OR: More air particles collide with the walls (1); Gas pressure increases inside (1); Balloon stretches (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Moves a stationary body: kicking a ball (1); Stops a moving body: catching a ball (1); Changes direction: hockey stick hits (1); Changes speed: pedalling faster (1); Changes shape: squeezing clay (1).	1+1+1+1+1

	OR: P = F/A (1); Larger force or smaller area gives greater pressure (1); 50/0.25 = 200 Pa (1); 50/0.5 = 100 Pa (1); Examples: knife, camel's feet (1).	1+1+1+1+1
27	Dissolving sugar (1); Spreading of ink (1); Smell of perfume (1); Spaces between particles (1); Particles in constant motion (1).	1+1+1+1+1
	OR: Composition (1); Separation method (1); Properties (1); Examples (1); Fixed ratio or not (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** newton	Named after Isaac Newton.
2	**(a)** pascal	1 Pa = 1 N/m squared.
3	**(b)** sugar dissolves in water without raising the water level much	Sugar particles occupy spaces between water particles.
4	**(d)** O_2	Oxygen contains atoms of only one kind.
5	**(b)** solute	Substance dissolved.
6	**(a)** lubricants	Oil forms a layer.
7	**(b)** wind	Wind is air in motion.
8	**(c)** evaporation	Surface process.
9	**(d)** ductility	Ductile.
10	**(a)** nitrogen	Most abundant gas.
11	**(d)** repel each other	Same charges repel.
12	**(d)** eye	Clear and calm.
13	**(c)** spread out in a windy place	More surface and wind.
14	**(c)** in a fixed ratio by mass	Law of constant composition.
15	**(d)** 20 per cent	20/100 x 100.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	To reduce friction (1); So the coins slide easily (1).	1+1
17	Pressure exerted by the weight of the air (1); Acts on everything on the Earth's surface (1).	1+1
18	Evaporation occurs at any temperature; boiling at a fixed temperature (1); Evaporation is at the surface; boiling throughout the liquid (1).	1+1
19	Element with properties between metals and non-metals (1); Silicon (1).	1+1
20	Solution is transparent, particles invisible; suspension is cloudy (1); Suspension particles settle and can be filtered (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Fastest particles escape from the liquid (1); Remaining particles have lower energy (1); Temperature of the liquid falls (1).	1+1+1
22	H and O are chemically combined (1); In a fixed ratio by mass (1); Water has properties different from H2 and O2 (1).	1+1+1
23	Mass of solute = 10/100 x 200 (1); = 20 g (1); Mass of water = 180 g (1).	1+1+1
24	Net force = 0 (1); Forces are balanced (1); The book stays at rest (1).	1+1+1
	OR: Sea breeze blows towards land (1); Land air rises, cooler air flows in (1); Pressure difference creates the wind (1).	1+1+1
25	Water vapour in air touches the cold glass (1); It loses energy and condenses (1); Droplets form (1).	1+1+1
	OR: Smallest unit that exists independently of a substance (1); Elements: O_2, H_2 (1); Compounds: H_2O, CO_2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Particle size (1); Appearance (1); Settling (1); Filtration (1); Examples (1).	1+1+1+1+1

	OR: Leaves hang together when uncharged (1); Touching with a charged rod gives charge to all parts (1); Leaves carry the same charge and repel (1); More charge, more divergence (1); Used to detect charge (1).	1+1+1+1+1
27	Measures air pressure (1); Falls with altitude (1); Low pressure signals storms (1); High pressure signals clear weather (1); Used in weather forecast (1).	1+1+1+1+1
	OR: Wrap cotton on the bulb (1); Dip in water, then in acetone (1); Note the temperature fall (1); Acetone gives more cooling (1); Faster evaporation leads to more cooling (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** the speed, direction or shape of an object	Effects of force.
2	**(d)** 50 Pa	100 / 2 = 50 Pa.
3	**(c)** constantly moving	Particles are always in motion.
4	**(a)** CO_2	Carbon dioxide has two kinds of atoms chemically joined.
5	**(a)** solvent	Dissolving medium.
6	**(b)** less	Rolling friction is smaller than sliding friction.
7	**(d)** differences in air pressure due to uneven heating	Air flows from high to low pressure.
8	**(c)** melting point	Melting point.
9	**(b)** sonority	Sonorous.
10	**(a)** suspension	Large particles.
11	**(c)** attract each other	Opposite attract.
12	**(d)** warm oceans	Needs warm water.
13	**(a)** cooling	Fastest particles leave, so cooling.
14	**(d)** mixture of gases	Mostly nitrogen and oxygen.
15	**(c)** 100 g	Solute + solvent.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Like charges repel (1); Unlike charges attract (1).	1+1
17	Calm, clear region at the centre (1); Surrounded by strong winds (1).	1+1
18	Condensation (1); Water vapour in air cools and becomes liquid (1).	1+1
19	It is soft and leaves a mark (1); It conducts electricity (1).	1+1
20	Mass of solution = 50 g (1); (5 / 50) x 100 = 10 per cent (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Drinks and tea (1); Medicines like ORS (1); Cleaning with soap solutions (1).	1+1+1
22	Static: pushing a heavy box that does not move (1); Sliding: a box sliding on the floor (1); Rolling: a ball rolling on the ground (1).	1+1+1
23	Sharp edge has small area (1); Same force gives larger pressure (1); Blunt edge spreads force over larger area (1).	1+1+1
24	Put a crystal of potassium permanganate in water (1); Colour spreads slowly (1); Particles move and mix on their own (1).	1+1+1
	OR: Generally dull (1); Brittle if solid (1); Poor conductors (except graphite) (1).	1+1+1
25	Add salt to water a little at a time and stir (1); Continue until no more dissolves (1); Undissolved salt settles: saturated (1).	1+1+1
	OR: Rubbing transfers charges from one body to another (1); One gains negative charge; other becomes positive (1); A glass rod rubbed with silk (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	H2: 2 hydrogen atoms (1); O2: 2 oxygen atoms (1); H2O: 2 H + 1 O (1); CO2: 1 C + 2 O (1); Molecule of an element vs compound (1).	1+1+1+1+1

	OR: Solid in liquid: sugar in water (1); Gas in liquid: soda water (1); Liquid in liquid: alcohol in water (1); Gas in gas: air (1); Solid in solid: brass (1).	1+1+1+1+1
27	Contact: muscular, friction (1); Non-contact: gravitational, magnetic (1); Electrostatic force acts at a distance (1); Contact forces need touching (1); Non-contact forces act through space (1).	1+1+1+1+1
	OR: Bottle with holes at different heights (1); Lowest stream goes farthest (1); Pressure increases with depth (1); A rubber-covered funnel in water shows pressure in all directions (1); Dams are thicker at the base (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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